

# Continuous DID

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## **Continuous Treatment**

Discrete multi-valued

Continuous valued dose

**Linear Model of without “Strong PTs”**

**Empirical Example**

# Setup

Individuals are observed in two periods ( $t = 0, 1$ )

Treatment turns on in the second period for some units

→ The untreated group receives no dose,  $D_i = 0$

→ The treated group receives a dose  $D_i > 0$

How do you do difference-in-differences in this setup?

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# Initial Example

Let's work through an example to illustrate what's novel in this setting...

There are a set of patients who sign up for an experimental medicine.

They are either not accepted (receive 0mg) or accepted into the program and allowed a positive dose

→ Doctor drugs 0mg, 10mg, 20mg

## Estimating effects for each dose group

We have three groups:

1. The 'untreated' group who received 0mg
2. The '10mg' group
3. The '20mg' group

The simplest thing to do is to treat the '10mg' and '20mg' group as two treatments and compare both to the 'untreated' group

# First-differences

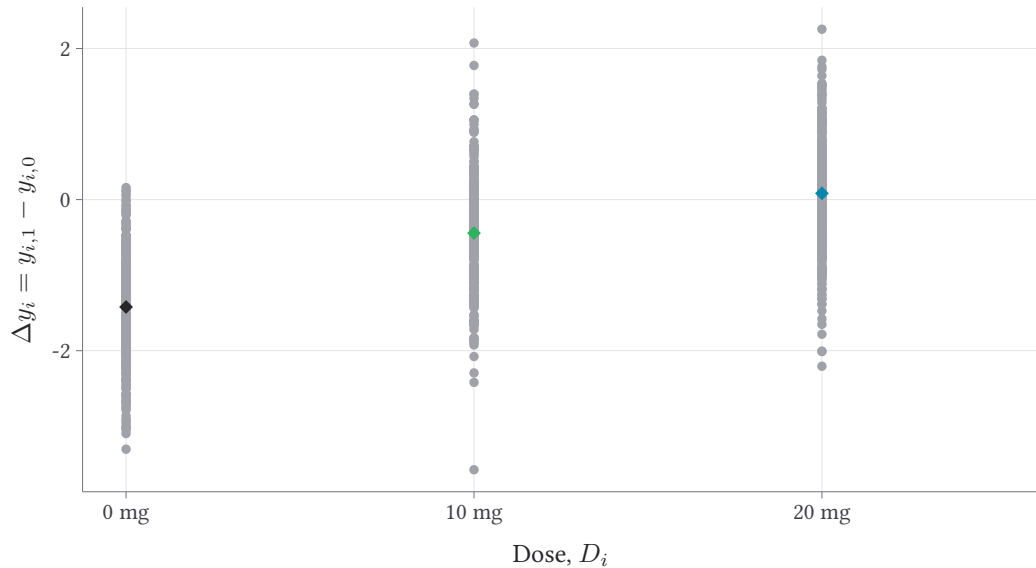
The rest of the slides will focus on *first-differences*:

$$\Delta y_i \equiv y_{i1} - y_{i0}$$

Parallel counterfactual trends assumption says that  $\Delta y_i$  would be on average the same for the '10mg' and the untreated group *in the absence of treatment*

→ Similarly when looking at the '20mg' group

◆ 0mg group avg.    ◆ 10mg group avg.    ◆ 20mg group avg.    ● Observations





# Diff-in-Diff

Our diff-in-diff estimate is formed as:

$$\hat{\tau}_{10} = \mathbb{E}[\Delta Y_i \mid D_i = 10] - \mathbb{E}[\Delta Y_i \mid D_i = 0]$$

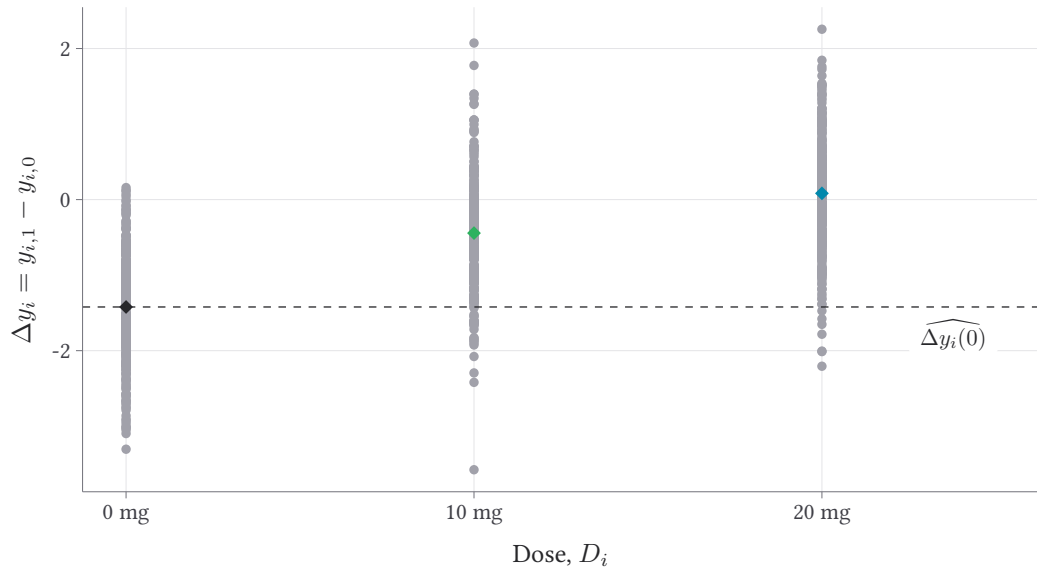
and

$$\hat{\tau}_{20} = \mathbb{E}[\Delta Y_i \mid D_i = 20] - \mathbb{E}[\Delta Y_i \mid D_i = 0]$$

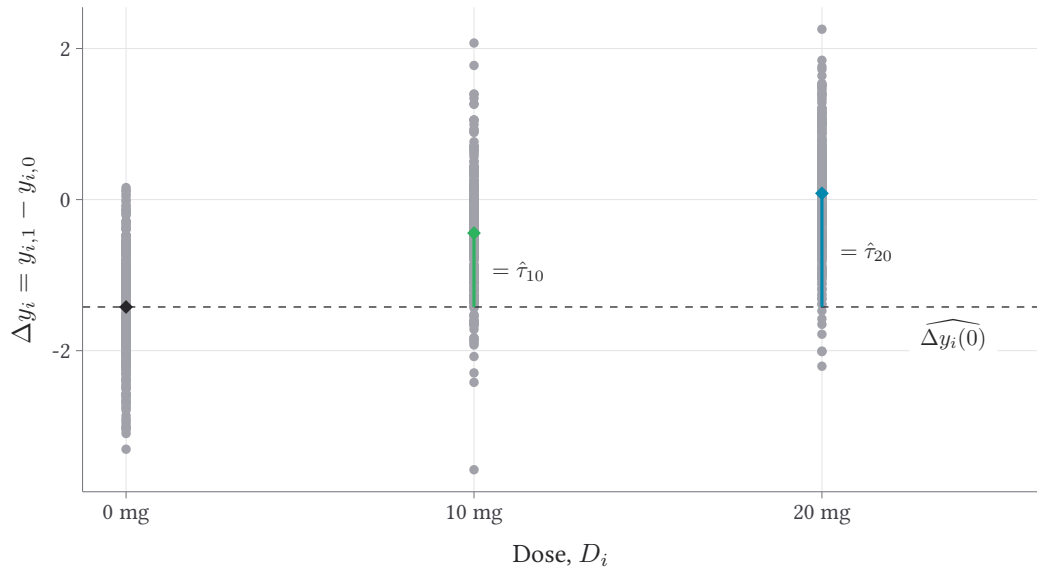
In both cases, we use the untreated group to estimate the counterfactual trend

→  $\mathbb{E}[\Delta Y_i \mid D_i = 0]$  is what we think the change in outcomes would have been in the absence of treatment for the treated unit

◆ 0mg group avg.    ◆ 10mg group avg.    ◆ 20mg group avg.    ● Observations



◆ 0mg group avg.    ◆ 10mg group avg.    ◆ 20mg group avg.    ● Observations



## Estimates

The previous figure shows two difference estimates of  $\hat{\tau}_{10}$  and  $\hat{\tau}_{20}$

→ The 20mg group has a higher estimated treatment effect than the 10mg group. What can we conclude from that?

# ATT Estimates

To understand this, let's remember the discussion of ATT vs. ATE

→ In the binary DID case, we estimate the average treatment effect *among those who received treatment*

Another way of writing the ATT estimate that will prove useful is

$$ATT \equiv \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1]$$

→ The  $\mid D_i = 1$  says we are averaging for the group whose treatment was  $D_i = 1$

# ATT Estimates

$$ATT \equiv \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1]$$

Why might the ATT be different than the ATE?

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The usual “selection on  $Y(1)$ ” story

→ Maybe units that benefit most from treatment select into it

# ATT Estimates

$$ATT \equiv \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1]$$

Why might the ATT be different than the ATE?

The usual “selection on  $Y(1)$ ” story

- Maybe units that benefit most from treatment select into it
- But, not parallel trends rule out a lot of the  $Y(0)$  stories
  - E.g. Units can not select into treatment because of negative shocks



## “ATT” in continuous DID

The concept of ATT vs. ATE is the same, but now there is no single ‘the treated’. Instead, there are two treated groups!

There are now three different potential outcomes

$$Y_{i1}(d),$$

for  $d = 0, 10, 20$

→ What the outcome would be had the unit taken 0mg, 10mg, or 20mg

→ We only observe one potential outcome  $Y_{i1}(D_i)$

## Treatment effects

We might care about the average effect of moving from 0 to some level  $d > 0$ :

$$Y_{i1}(d) - Y_{i1}(0)$$

Or, we might want to know about the effect of increasing someone's dose from  $d \rightarrow d'$

$$Y_{i1}(d') - Y_{i1}(d)$$

→ E.g. doctors wonder about increasing someone's dose to the next level

## What did our two diff-in-diff estimates identify?

So what do we call the 'ATT'?  $ATT_{10}$  and  $ATT_{20}$  perhaps?

Callaway, Goodman-Bacon, and Sant'Anna call this:

$$ATT(d \mid d) = \mathbb{E}[Y_{i1}(d) - Y_{i1}(0) \mid D_i = d]$$

→ Our diff-in-diff's can only estimate the effect of a certain dose *among the units that received that dose*.

$$ATT(d \mid d)$$

These dose-specific ATTs are identified from the same parallel trends assumption we are used to:

→ Had the units receiving dose  $D_i = d$  not received any dose  $D_i = 0$ , their trend in  $Y$  would have evolved the same as the untreated units.

The good news. We know how to think about the parallel trends assumption

→ can potentially assess this assumption using pre-trends placebo checks

# Comparing estimates

We observe a set of  $ATT(d | d)$  for a set of doses (in our example 10 and 20)

The initial temptation is to compare these two estimates:

- E.g. finding the ‘correct’ dose of a medicine by picking the highest ATT
- E.g. finding diminishing returns on an treatment investment
- E.g. what is the ‘per-unit’ effect of the treatment?

# Thinking about counterfactual treatment effects

We want to think about what would happen if one group of units received a *different* dose?

One example is what Callaway, Goodman-Bacon, and Sant'Anna call the **Average Causal Response**,  $\text{ACRT}(d' \mid d)$ :

Effect of increasing dose by a little

$$\text{ACRT}(d' \mid d) = \mathbb{E} \left[ Y_{i1}(d' + \varepsilon) - Y_{i1}(d') \mid D_i = d \right]$$

For the group that received  $d$

## Average Causal Response

$$\text{ACRT}(d' \mid d) = \mathbb{E} \left[ Y_{i1}(d' + 1) - Y_{i1}(d') \mid D_i = d \right]$$

- This measures the ‘per-unit’ effect of the treatment (at some starting point  $d'$ )
- The effect is averaged over the group of units that received treatment  $D_i = d$

## Average Causal Response

$$\text{ACRT}(d' \mid d') = \mathbb{E} \left[ Y_{i1}(d' + 1) - Y_{i1}(d') \mid D_i = d' \right]$$

Could imagine estimating this using our two ATT( $d \mid d$ ):

$$\text{ACRT}(d_1 \mid d_1) \approx \frac{\text{ATT}(d_2 \mid d_2) - \text{ATT}(d_1 \mid d_1)}{d_2 - d_1}$$



# Thinking about counterfactual treatment effects

Note that two things change when we move from  $d_1$  to  $d_2$

- The dose amount they receive
- And the group of units we are estimating ATTs for

The latter is what creates difficulty:

- If the group of units receiving treatment at  $d_1$  is different than the group receiving treatment at  $d_2$ , then the ATTs are not directly comparable

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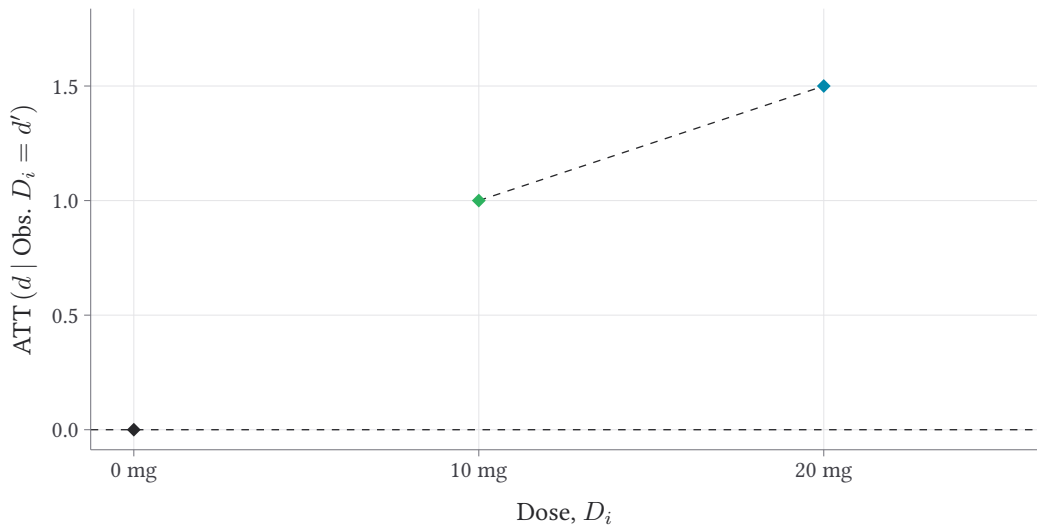
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Let's dig into this out with our example...

## Estimated ATTs

◆ 10mg group    ◆ 20mg group



# Homogeneous dose-ATTs

On the one extreme, we have homogeneity of treatment effects:

$$ATT(d' \mid d) = ATT(d')$$

→ Treatment effects can depend on dose amount  $d'$ , but not on which treatment dose-group you are looking at

# Homogeneous dose-ATTs

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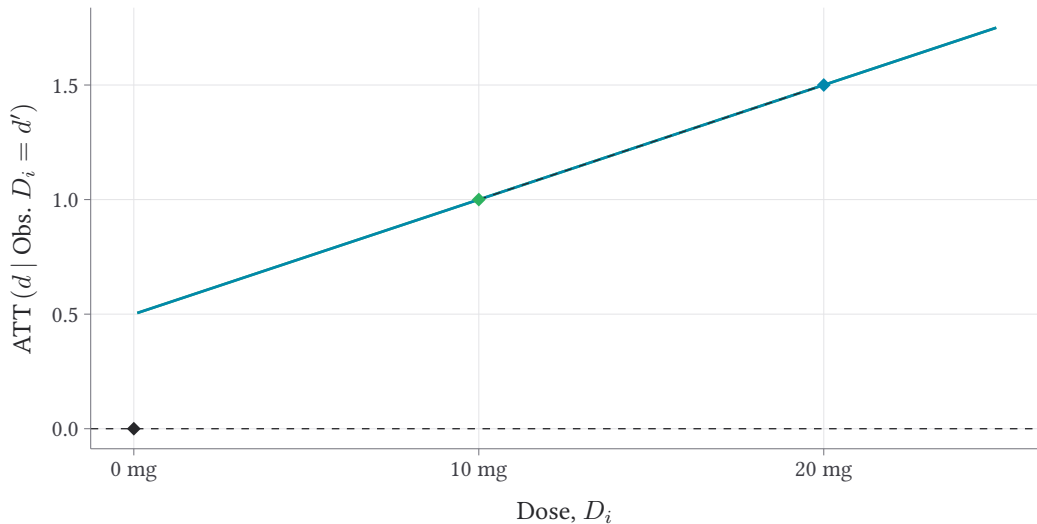
→ Treatment effects can depend on dose amount  $d'$ , but not on which treatment dose-group you are looking at

This let's you directly compare  $ATT(d' | d')$  and  $ATT(d | d)$  to estimate

$$ACRT(d' | d) = ACRT(d')$$

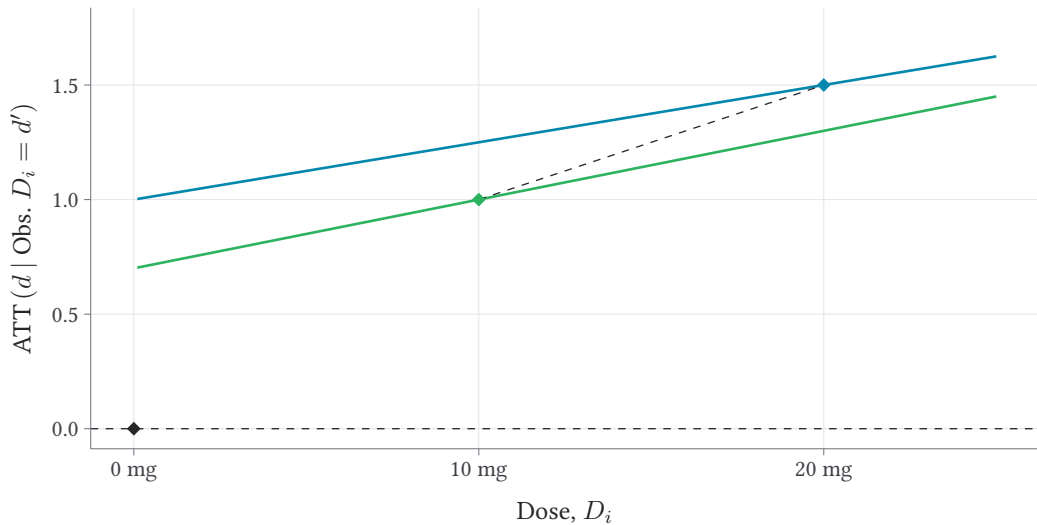
# Homogeneous and Linear Dose-response, ATT ( $d \mid d'$ )

◆ 10mg group    ◆ 20mg group



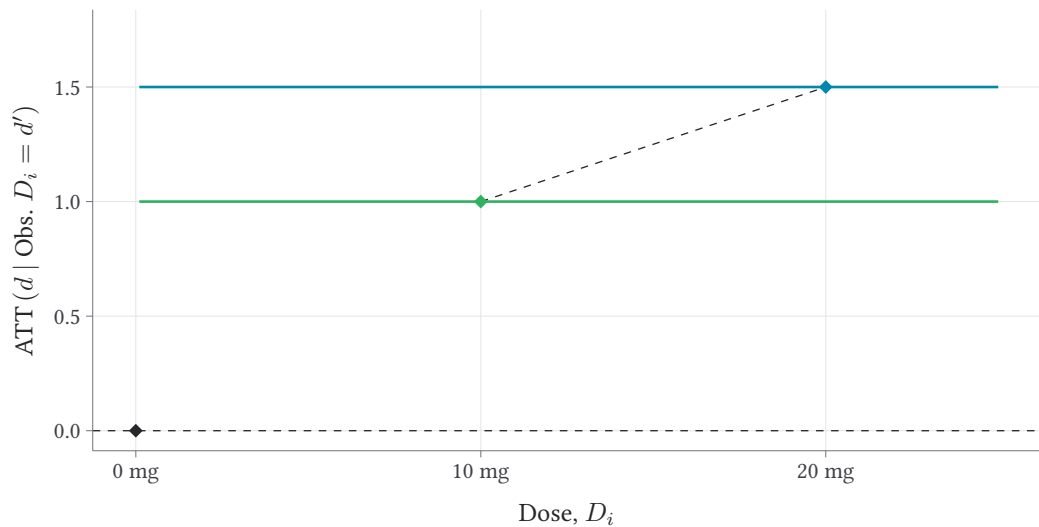
# Heterogeneous Linear Dose-response

◆ 10mg group    ◆ 20mg group



## Constant Dose-response

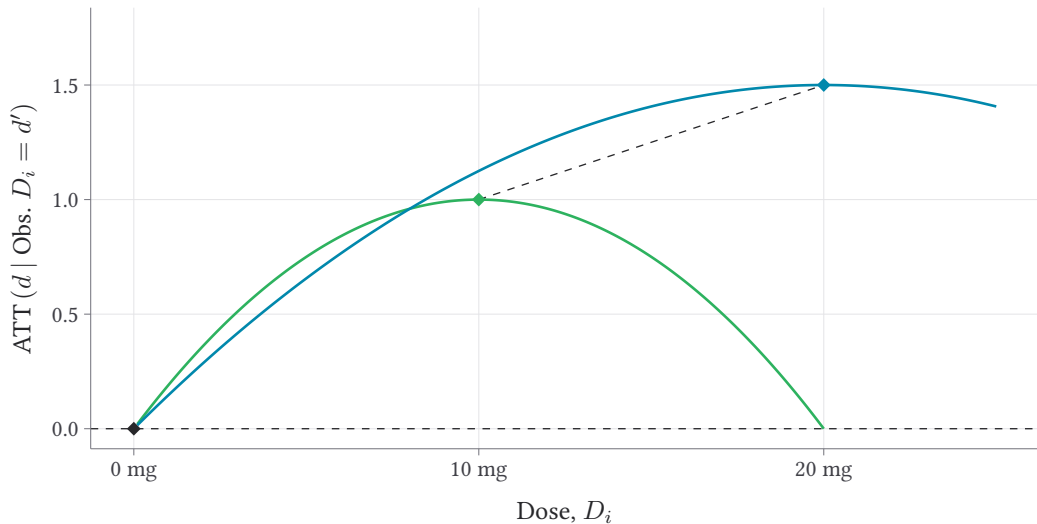
◆ 10mg group    ◆ 20mg group





## People select dose that maximizes their outcome

◆ 10mg group    ◆ 20mg group



# Recap

So as a recap:

1. Dose-group specific DID estimates identify the  $ATT(d \mid d)$  effects
2. Comparing across different doses does not inform us of the marginal effect of changing a dose, the  $ACR(d \mid d)$

# “Strong Parallel Trends”

In Callaway, Goodman-Bacon, and Sant’Anna, they refer to an assumption called “Strong Parallel Trends” which combines two assumptions:

1. The parallel counterfactual trends assumption: each dose group has the same counterfactual trends as the untreated group
2. Treatment effect homogeneity across groups:  $ATT(d | d) = ATT(d)$ 
  - $\Rightarrow ACR(d | d) = ACR(d)$

The latter is only needed to identify the ACR curve using your  $ATT(d | d)$  estimates

## “Strong Parallel Trends”

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The first assumption is what allows estimation of  $ATT(d | d)$

→ DID of units with  $D_i = d$  compared to  $D_i = 0$  group

## “Strong Parallel Trends”

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The first assumption is what allows estimation of  $ATT(d | d)$

→ DID of units with  $D_i = d$  compared to  $D_i = 0$  group

The latter assumption is what let's us compare  $ATT(d)$  across  $d$  to identify the ACR curve

→ With continuous dosage, the slope of  $ATT(d)$  is the ACR curve

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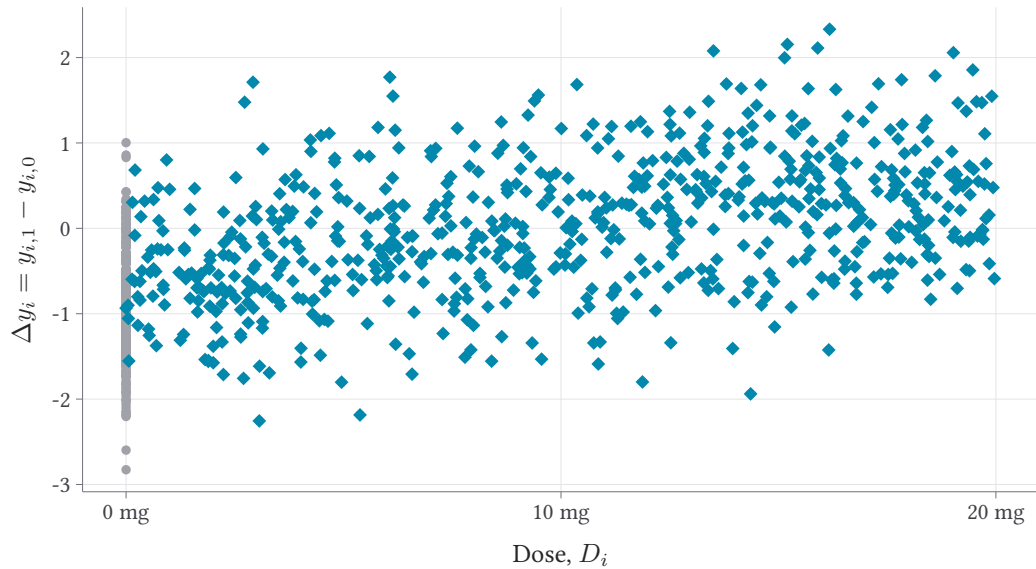
**Empirical Example**

## Doctor example extended

Now, let's say we have a continuous distribution of doses

- Some units receive 0mg
- Others receive some amount between 0 and 20mg

● Observations    ◆ Dose group avg.

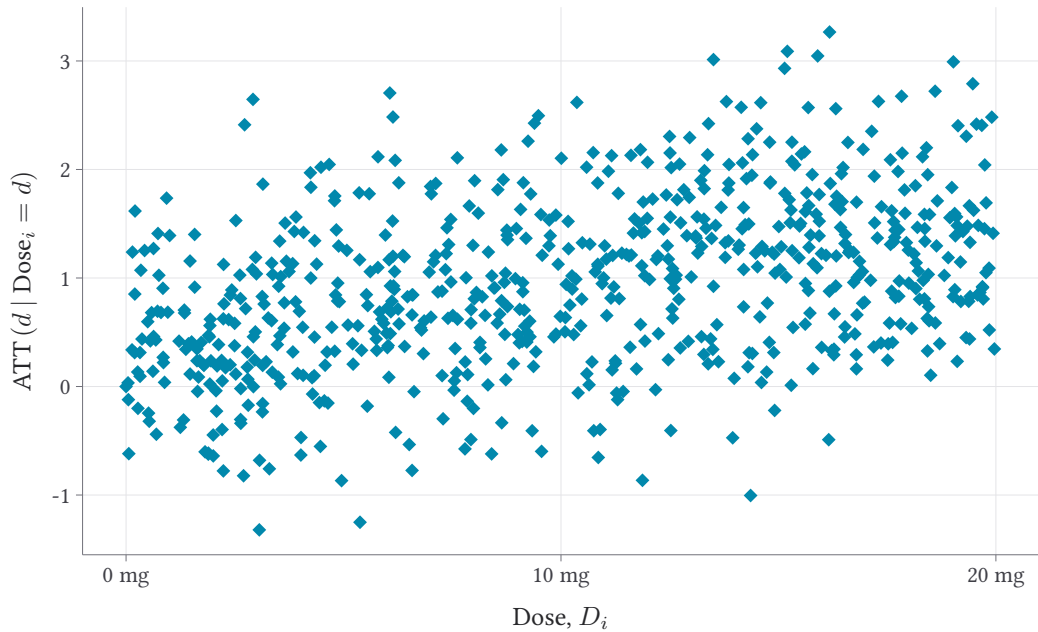




## Dose-specific DID estimates

Our previous method of having dose-specific DID estimates will be quite noisy in this setting

- Very few individuals with the *same* dose
- Throwing out information of individuals with very similar dose



## Alternative strategies

Our previous strategy clearly will not suffice

An alternative strategy is to *pool* observations with similar doses to estimate an average

$$\mathbb{E}[\Delta Y_i \mid D_i = d]$$

- E.g. say you break doses into 2mg bins (0–2, 2–4, etc.) and take averages in those bins
- Or any other of your favorite non-parametric method for estimating

## ATT( $d \mid d$ ) estimators

In general, our ATT( $d \mid d$ ) estimators will take the form of

$$\Delta Y_i - \hat{\mathbb{E}}[\Delta Y_i \mid D_i = 0]$$

- The second term is just the average change in  $Y$  from the control group
- The first-term is where we have some options

## Option 1: Overall ATT

For all treated units, calculate

$$\Delta Y_i - \hat{\mathbb{E}}[\Delta Y_i \mid D_i = 0]$$

→ It is the *observed* change in  $Y$  minus the counterfactual (estimated) change in  $Y(0)$

Then, average  $\Delta Y_i - \hat{\mathbb{E}}[\Delta Y_i \mid D_i = 0]$  for everyone with  $D_i > 0$

→ This approach is unsatisfying because we've effectively 'binarized' our treatment into a simple DID setup

## Option 2: Bins

Instead, we can use a classic trick of ‘discretizing’ the dose into a set of bins

→ Similar to using age bins instead of polynomial of age

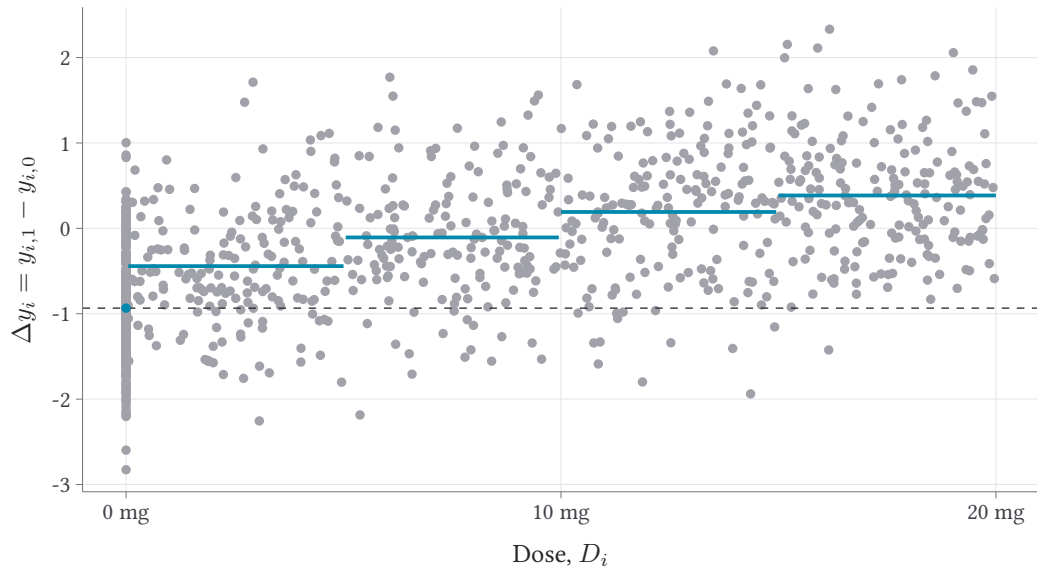
## Option 2: Bins

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→ Similar to using age bins instead of polynomial of age

Take  $\Delta Y_i - \hat{\mathbb{E}}[\Delta Y_i \mid D_i = 0]$  and regression on indicators for each bin

● Observations    — Dose bins avg.





# Pros and Cons of Bins

## Advantages:

- Creates a discrete set of groups that are easy to discuss in your paper
- Event-studies are easy to estimate (one for each bin)

## Disadvantages:

- Bins create a flat surface that is likely not perfectly specified (but gets close as the number of bins grow)
- Poor estimates of  $ACRT(d)$  because the slope is 0 everywhere except the bin endpoints.

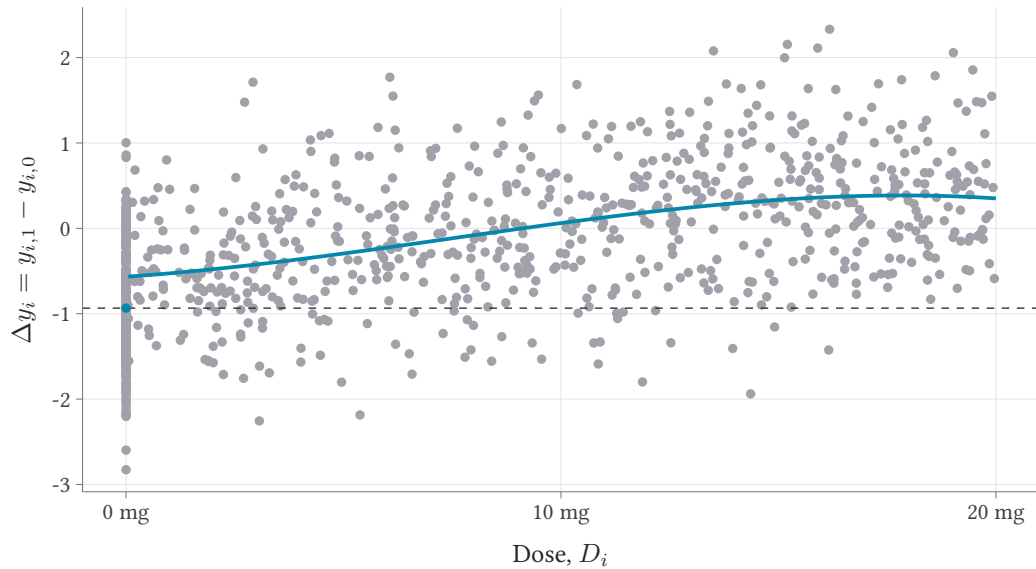
## Option 3: Non-parametric smoothing

Alternatively,  $\hat{\mathbb{E}}[\Delta Y_i \mid D_i = d]$  can be estimated with non-parametric methods

The simplest way is to use non-parametric method on  $\Delta Y_i - \hat{\mathbb{E}}[\Delta Y_i \mid D_i = 0]$  with only the  $D_i > 0$  observations.

- The result of this will be the DID estimates
- Standard errors produced should be correct (can always bootstrap to be safe)

● Observations    — Smoothed Average of  $(\Delta y_i)$



## contdid

The 'contdid' package by CGBS does this directly non-parametric smoothing method (trimmed source):

```
m0 <- mean(dy[dose == 0])  
dy_centered <- dy - m0  
  
## Non-parametric smoothing estimator  
cck_res <- npiv::npiv(  
  Y = dy_centered[dose > 0],  
  X = dose[dose > 0],  
  knots = "quantile"  
)
```

# Pros and Cons of Non-parametric smoothing

## Advantages:

- Better estimates the smooth evolution of  $\hat{\mathbb{E}}[\Delta Y_i \mid D_i = d]$

## Disadvantages:

- Event-studies are hard to display (one plot for each event-time?)
- Pre-trends are hopefully flat across doses
  - Only assesses half of the strong parallel trends assumption
- The ACRT( $d$ ) estimate is the derivative of the smoothed curve (can estimate using spline procedures)

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## **Empirical Example**

## Linear in Dose

Consider the standard TWFE type model (written in first-differences):

$$\Delta Y_i = \alpha + \tau D_i + u_i$$

- The unit-fixed effect cancels out
- $\alpha$  is the common time-trend (the  $D_i = 0$  group will estimate)
- $D_i$  is assumed linear with common coefficient

# OLS Coefficient

The (true) heterogeneous treatment effects are given by  $ATT(D_i | D_i)$ .

$$\Delta Y_i - \hat{\alpha} = ATT(D_i | D_i) + \text{noise} = \tau D_i + u_i$$

Our regression coefficient is the best linear predictor of the data  $y = ATT(D_i | D_i) + \text{noise}$  and  $x = D_i$



# OLS Coefficient

$$\Delta Y_i - \hat{\alpha} = \text{ATT}(D_i \mid D_i) + \text{noise} = \tau D_i + u_i$$

OLS compares units with larger to smaller doses. We might be *really tempted* to call  $\hat{\tau}$  the slope the dose-specific ATT curve When doing so, it does two things:

1. changes the dose received
2. changes the group of treated units you're examining

## Bias of coefficient for ACRT

$$\Delta Y_i - \hat{\alpha} = \text{ATT}(D_i \mid D_i) + \text{noise} = \tau D_i + u_i$$

OLS compares units with larger to smaller doses. When doing so, it does two things:

1. changes the dose received
2. **changes the group of treated units you're examining**

It is the second change that prevents us from estimating the effect of increasing the dose for units (marginal effect)

## Adding ‘Strong PTs’

$$\Delta Y_i - \hat{\alpha} = \text{ATT}(D_i) + \text{noise} = \tau D_i + u_i$$

If we assume strong parallel trends, then  $\hat{\tau}$  estimates a weighted average of the  $ACR(d)$  curve

→ Something like an “average slope”  $\text{ATT}(D_i)$

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## **Empirical Example**

This example will use data from “Trade Liberalization and Markup Dispersion: Evidence from China’s WTO Accession” (Lu and Yu, American Economic Journal: Applied Economics 2015)

# China's WTO Ascension

The paper uses the entrance of China into the World Trade Organization (WTO) in 2002

When entering the WTO, tariffs at the industry level were expected to drop to at most 10%

## Defining continuous treatment variable

The research strategy is to compare industries (at the 3-digit SIC level) that had higher or lower pre-existing tariff rates

- After the entrance in 2002, industries with higher tariff had *larger* decreases in tariffs
  - ⇒ a larger shock in import competition

Can define treatment as  $D_i = \max(0, \text{Tariff}_{2001} - 0.10)$

- How much tariffs are expected to drop

*Aside: Often reframing a continuous variable (Tariff) to changes (Change in Tariff) can help you have a  $D_i = 0$  group*

## Outcome variable: Markup Dispersion

The dataset is an industry-by-year panel dataset

→ Each row is an industry in China in a given year

The outcome variable is the industry-level *Theil Markup Index*, which measures how much spread in markups at the firm level

→ Higher markup index  $\Rightarrow$  less competitive market



## Outcome variable: Markup Dispersion

The dataset is an industry-by-year panel dataset

→ Each row is an industry in China in a given year

The outcome variable is the industry-level *Theil Markup Index*, which measures how much spread in markups at the firm level

→ Higher markup index  $\Rightarrow$  less competitive market

**Hypothesis:** Larger decrease in tariffs  $\Rightarrow$  more competition  $\Rightarrow$  decrease in markup dispersion

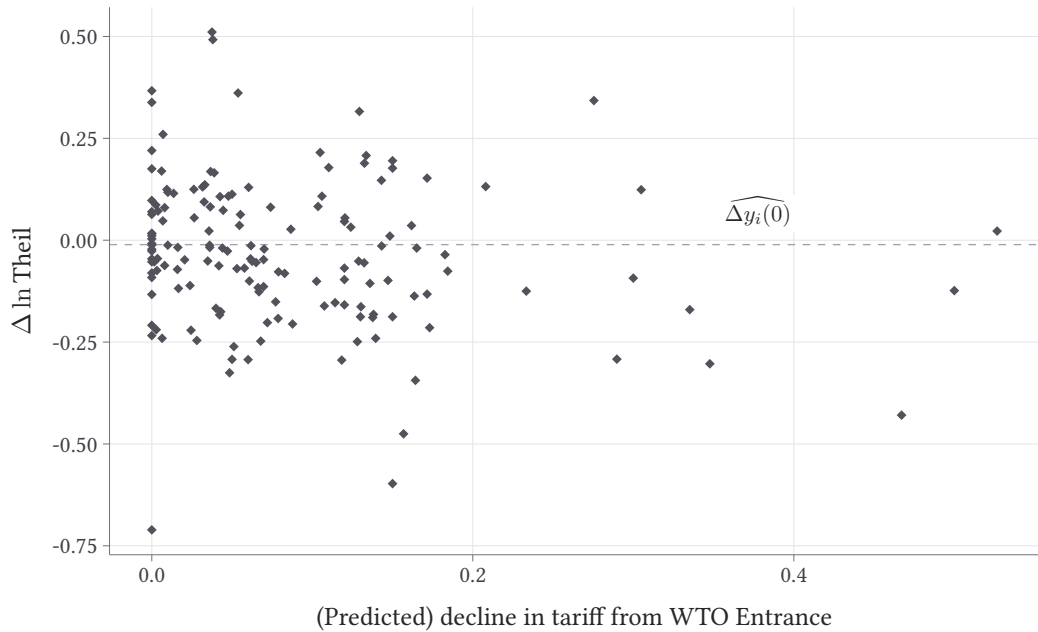
## $2 \times 2$ setup

We will use 2000 ( $t = 0$ ) and 2004 ( $t = 1$ ) as our initial years

Let's look at the raw data

→  $x$ -axis is our treatment variable  $D_i = \max(0, \text{Tariff}_{2001} - 0.10)$

→  $y$ -axis is the change in markup dispersion  $\Delta \ln \text{Theil}_i = \ln \text{Theil}_{i,2004} - \ln \text{Theil}_{i,2000}$



## Parallel Trends assumption

Our identification assumption is that if China did not enter the WTO, counterfactual changes in markup dispersion is the same for the unaffected group and the affected doses

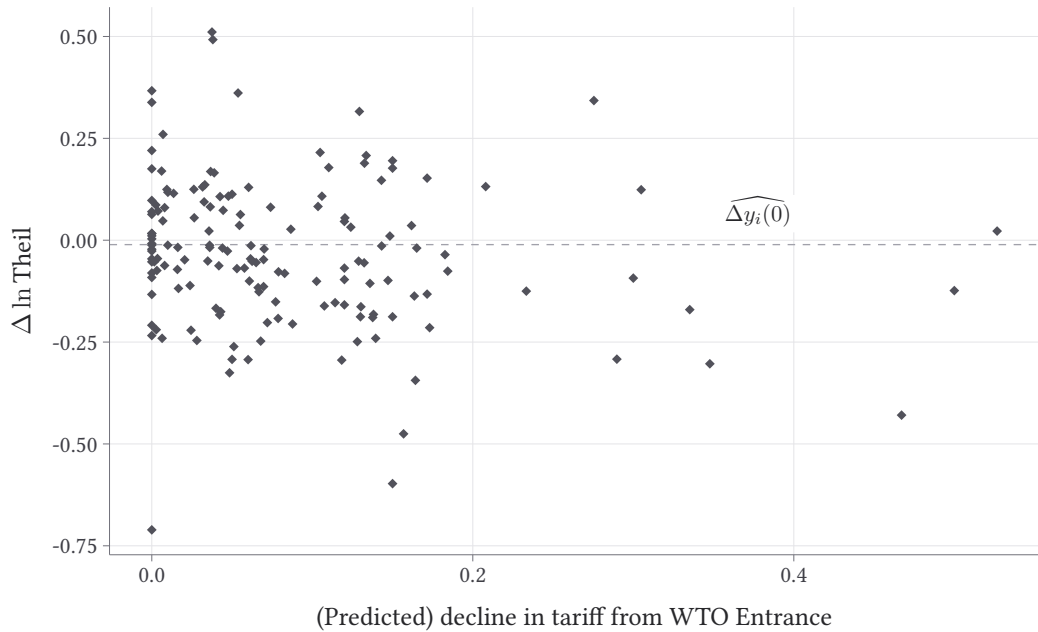
→ We can assess the plausability using pre-periods (later)

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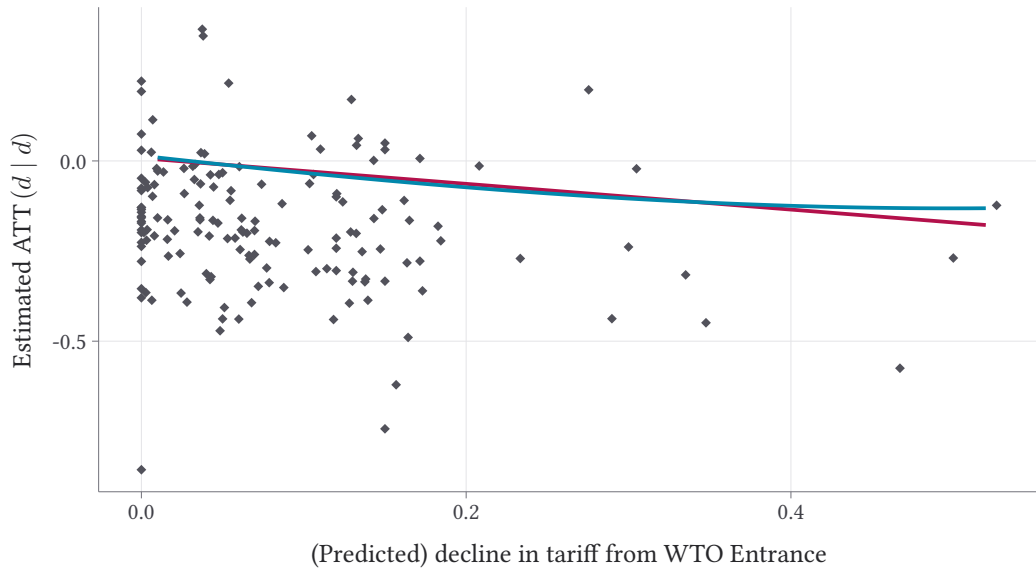
Our identification assumption is that if China did not enter the WTO, counterfactual changes in markup dispersion is the same for the unaffected group and the affected doses

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To estimate counterfactual trend, we will take average  $\Delta \ln Theil_i$  for the  $D_i = 0$  group



Linear Estimate    Spline Estimate



## Pre-trends ‘test’

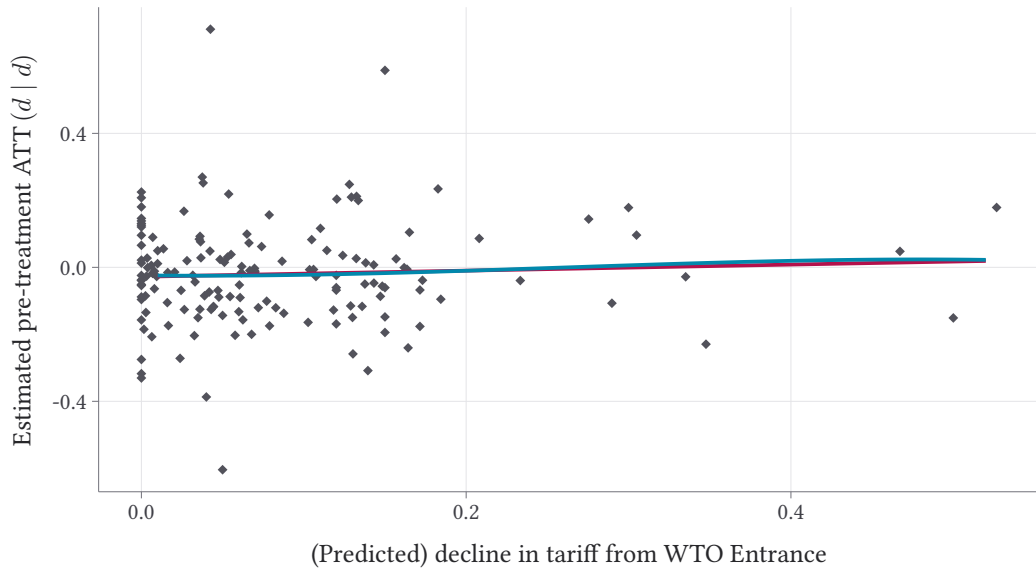
Redo the previous estimates using

$$\Delta \ln \text{Theil}_i = \ln \text{Theil}_{i,1998} - \ln \text{Theil}_{i,2000}$$

We are hoping that the pre-trend change in markup dispersion is uncorrelated with the dose received,  $D_i$



Linear Estimate    Spline Estimate



## Adding covariate-specific trends

If we want to assume that units with the same value of  $X_i = x$  are on the same counterfactual trend (regardless of their  $D_i$ ), we can!

1. Estimate  $\Delta Y_i = X_i\gamma + v_i$  using  $D_i = 0$  group (or a more flexible function of  $X_i$ )
2. Form  $\Delta Y_i - X_i\hat{\gamma}$  where  $X_i\hat{\gamma}$  is the estimated  $\Delta Y_i(0)$  for the treated units
3. Regress  $\Delta Y_i - X_i\hat{\gamma}$  on your functions of  $D_i$  (linear, bins, splines)